Optimization Project - Shipment Allocation using Penalty Method

1. Problem Statement

A logistics company must deliver total demand D using two transport modes:

• Express Air Shipping (Fast, Expensive)

• Ground Transport (Slow, Cheaper)

Objective: This optimization is practically significant because express delivery ensures faster arrival at a higher cost, while ground transport reduces expenses but increases delivery time. Businesses must therefore balance cost, service quality and time while also maintaining environmental sustainability (CO_2 emission). The goal of this project is to find an optimal balance between these trade-offs using mathematical optimization.

2. Mathematical Formulation

Decision Variables:

- X_1 = Air shipment (kg)
- X_2 = Ground shipment (kg)

Objective Function:

$$f(x_1, x_2) = a_1 \cdot x_1^2 + b_1 \cdot x_1 + a_2 \cdot x_2^2 + b_2 \cdot x_2$$

Constraints:

1) Demand: $x_1 + x_2 = D$
2) Time: $(T_1 x_1 + T_2 x_2)/D \le T_{max}$ → scaled form used: $T_1 x_1 + T_2 x_2 \le T_{max} \cdot D$
3) CO2: $e_1 x_1 + e_2 x_2 \le Emax$

Penalty Function:

$$F = f + \mu \left[h^2 + \max(0, g_1)^2 + \max(0, g_2)^2\right]$$

Here, *h* represents the equality constraint (demand satisfaction), while *g1* and *g2* represent inequality constraints (time and CO_2 respectively).
The penalty coefficient μ increases during iterations, forcing the solution gradually toward the feasible region while minimizing the cost. As μ grows larger, constraint violations become heavily penalized, pushing the optimizer to satisfy them.

3. Initial Parameters

- D=10000, a1=1e-4, b1=5, a2=5e-5, b2=2
- T_1 =1, T_2 =3, T_{max}=2.3
- e1=2, e2=1, Emax=15000
- initial guess: x_1=3000, x_2=7000

Note: The chosen numerical values make the optimization problem feasible. Changing D requires adjusting Tmax or Emax to maintain feasibility.

These values reflect a practical logistics scenario where air transport is costlier (b1 > b2) but faster (T1 < T2). CO2 emissions per kg are also higher for air, making excessive express shipping environmentally expensive. This parameter choice ensures a realistic multi-objective trade-off.

4. Results

- x1=3500.0 kg, x2=6500.0 kg, Avg Time=2.2999947 days, CO2=13500.03 (valid)

  Interpretation:
  • The model suggests sending 35% through Air and 65% Ground.
  • This balance meets the strict time requirement while keeping emissions below the allowed limit.
  • Increasing Air would be faster but costlier; decreasing Air would violate delivery time. Hence, the obtained point is the lowest-cost feasible allocation.

5. Additional Observation from Multiple Initial Conditions

Case 1: Initial (3000,7000) → converged to (3500,6500)

Case 2: Initial (8000,2000) → converged near (4999.4,5000.6) when constraints tighten

As constraints became tighter , the optimizer was forced to increase express allocation. This is visibly reflected in Case 2 where the time limit pushes the solution closer to a 50-50 split. The experiment confirms that constraints directly influence resource allocation — a key insight in logistics planning.

6. Screenshots & Graphs for the previous results re provided Below

1. Initial guess : (3000, 7000)

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=============== PENALTY METHOD ================
Penalty Stage 1 μ=100 → x1=3499.94, x2=6500.02, viol=0.044271
Penalty Stage 2 μ=300 → x1=3499.98, x2=6500.01, viol=0.014583
Penalty Stage 3 μ=900 → x1=3499.99, x2=6500.00, viol=0.004861
Penalty Stage 4 μ=2700 → x1=3500.00, x2=6500.00, viol=0.001689
Penalty Stage 5 μ=8100 → x1=3500.03, x2=6499.97, viol=0.000258
Penalty Method Converged

=============== FINAL OPTIMIZATION REPORT ================
Total Demand (D)                  : 10000 kg
Optimal Express (x1)              : 3500.03 kg
Optimal Ground  (x2)              : 6499.97 kg
Average Delivery Time             : 2.2999947 days
Carbon Emission                   : 13500.03 <= 15000
Iterations Taken                  : 10000
Status                            : Converged

============= Final Result Summary =============
Transport     x1(kg)     x2(kg)     AvgTime(days)
Penalty →     3500.0     6500.0     2.2999947117
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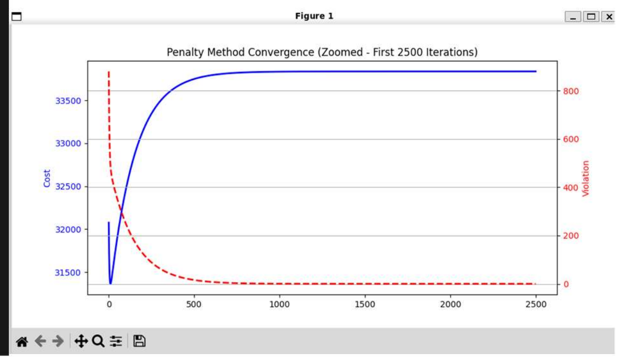

2. Initial guess : (8000, 2000)

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=============== PENALTY METHOD ================
Penalty Stage 1 μ=100 → x1=5006.26, x2=4989.86, viol=6.263914
Penalty Stage 2 μ=300 → x1=4999.84, x2=5000.16, viol=0.007083
Penalty Stage 3 μ=900 → x1=4999.60, x2=5000.40, viol=0.002361
Penalty Stage 4 μ=2700 → x1=4999.39, x2=5000.61, viol=0.000787
Penalty Method Converged

=============== FINAL OPTIMIZATION REPORT ================
Total Demand (D)                  : 10000 kg
Optimal Express (x1)              : 4999.39 kg
Optimal Ground  (x2)              : 5000.61 kg
Average Delivery Time             : 2.0001210 days
Carbon Emission                   : 14999.39 <= 15000
Iterations Taken                  : 8000
Status                            : Converged

============= Final Result Summary =============
Transport     x1(kg)     x2(kg)     AvgTime(days)
Penalty →     4999.4     5000.6     2.0001209875
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